Bin Packing -- Container Loading

Generated by uber-polya

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find

Domain: Integer Linear Programming

Universe

- See problem description

Variables

Symbol	Meaning	Type / Range
x	decision variable	\mathbb{{R}}

Structure

- **Items**: 25 packages with weights ranging from 5 to 45 kg (generated from seed=42)
- **Container capacity**: 100 kg each
- **Objective**: Minimize the number of containers used
- **Approaches**: First Fit Decreasing (heuristic) vs. ILP (exact optimal)

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

(1) Bin packing**: Classic NP-hard combinatorial optimization problem

(2) First Fit Decreasing (FFD)**: Sort items largest-first, place each in the first bin that fits

(3) NP-hardness**: No known polynomial-time exact algorithm; heuristics provide fast approxi

(4) Approximation ratio**: FFD uses at most ceil(11/9 * OPT + 6/9) bins

(5) ILP relaxation**: Integer Linear Programming finds the provably optimal solution

Approach

Suggested approach: FFD heuristic + ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: FFD heuristic + ILP

2. Solution

Answer:	Solution computed
Optimal:	Yes
Feasible:	Yes
Algorithm:	FFD heuristic + ILP (PuLP/CBC, Branch & Bound)
Time:	0.0292s
Certificate:	Branch-and-bound solved to optimality.

Solution Details

Solution computed successfully.

Verification

✓	Feasibility	All constraints satisfied
✓	Optimality	FFD heuristic + ILP (PuLP/CBC, Branch & Bound)

3. Interpretation

The Question

- **Items**: 25 packages with weights ranging from 5 to 45 kg (generated from seed=42)
- **Container capacity**: 100 kg each
- **Objective**: Minimize the number of containers used
- **Approaches**: First Fit Decreasing (heuristic) vs.

The Answer

A optimal solution was found.

What This Means

- FFD heuristic: 7 bins, average utilization ~93%
- ILP optimal: 7 bins, average utilization ~93%
- Lower bound: ceil(648 / 100) = 7 bins
- Utilization analysis: Both FFD and ILP achieve the lower bound; FFD minimum bin utilization is 59% while ILP achieves at least 83%
- All 8 verification checks pass

Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio